主题：机器学习预测收益的不确定性

- 机器学习已被广泛用于风险溢价预测。（Gu et al.,2020）
- 现有文献仅关注“点预测”，即根据收益预测大小构建 H-L 组合
- 这隐含了强假设：高预测收益 = 高投资价值。然而，若预测本身不精确，这种排序可能并不可靠。
  - 事实上，传统文献早已指出，风险溢价估计本身就具有较强不确定性（Fama and French, 1997; Pástor and Stambaugh, 1999）
  - 对于参数更多、结构更复杂的机器学习模型，同理也存在不确定性性
- 投资者不仅要关心“预测收益有多高”，还应关心“这一预测有多可信”；将预测值与预测精度结合，可能改进投资组合构建。

- 主题：机器学习预测收益的不确定性

- 重点：如何度量不确定性/置信度?
  - 机器学习的贝叶斯解释--计算后验分布作为不确定性 (Allena，2025)
  - closed-form OLS 形式；bootstrap 计算预测误差。(Liao et al.，2026)

Confident Risk Premiums and Investments Using Machine Learning Uncertainties

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(RFS,2025)

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2026 年 4 月 12 日

Question

- Can ML forecasts become more useful for invest when account for uncertainty?
  - How to measure Machine Learning Uncertainties?
    - Bayesian + ML(linear, NN)
  - Do stocks with high predicted risk premia and high forecast confidence generate better return?
    - Yes
    - Model more complex, gains larger.

Why interesting?

- **Background:** Machine learning is now widely used to predict risk premia.
- **Limitation:** Existing studies mainly focus on **point forecasts**, while paying little attention to how **precise** these forecasts are.
- Risk premia are inherently difficult to estimate precisely (Fama and French, 1997), and this issue may be even more important for complex ML models.
- **Method gap:** While confidence intervals are well studied in linear models, they are much harder to obtain for ML. → This paper： Bayesian approach.
- **Investment implication:**
  - A stock with high predicted return but a wide confidence may not be attractive.
  - **High return + high confidence** may lead to better portfolio performance.

Contribution

- **Machine learning and asset pricing literature**
  - prior: construct portfolios based solely on return forecasts(Lewellen,2015)
  - extend: forecast confidence intervals contain economically valuable information.
- **Machine learning estimate variances literature**
  - Gal and Ghahramani (2016) assume an i.i.d. model specification in ML area.
  - first to estimate variances for risk premium forecasts based on penalized linear models and NNs

Design–Measure ex-ante forecast confidence

- **Challenge:** ML models usually deliver point forecasts, but investors also need to know how precise those forecasts are.
- **Core:** Measure ex-ante confidence–The paper defines ex-ante confidence as:

$$EC_{it} = \frac{\left|\hat{E}_t(r_{i,t+1})\right|}{\sqrt{\widehat{Var}\left(\hat{E}_t(r_{i,t+1})\right)}} \tag{1}$$

  - A forecast is more confident when the predicted mean is large relative to its forecast standard error.

- **Approach:** Derive the forecast variance of each stock's predicted risk premium using Bayesian posterior distributions.

Design–Measure ex-ante forecast confidence

残差非i.i.d（贝叶斯假设）

$$r_{i,t+1} = z_{i,t}^{\top}\beta + \eta_{i,t+1}$$

特征

残差非独立，受到共同因子冲击f_{t+1}:

$$Mkt_{t+1} - E_t(Mkt_{t+1}))$$

$$\eta_{i,t+1} = \Lambda_{i,t} f_{t+1} + \varepsilon_{i,t+1}$$

股票i对冲击的敏感度：

$$\Lambda_{i,t} = a_0 + a_1 Size_{i,t} + a_2 BM_{i,t} + a_3 Mom_{i,t}$$

未知，指定bayes先验

ML的贝叶斯解释

线性模型： $\hat{\beta} = \arg\min_{\beta} \frac{1}{N^{Tr}N^{S}} \sum_{t\in Tr}\sum_{i\in S}\left(r_{i,t+1} - z_{i,t}^{\top}\beta - \Lambda_{i,t}f_{t+1}\right)^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2$

定理1：当设置β先验为下式时， β的后验分布等价于线性模型的最小化MSE(附录证明)

$$\pi(\beta \mid \lambda_1, \lambda_2, \sigma_\epsilon^2) \propto \exp\left\{-\frac{\lambda_1}{\sigma_\epsilon}\sum_{j=1}^{P}|\beta_j| - \frac{\lambda_2}{2\sigma_\epsilon^2}\sum_{j=1}^{P}|\beta_j|^2\right\}$$

设置先验：β，σ，λ，α → 估计后验：训练样本(z,r)+ Gibbs 抽样 → 估计Var($\widehat{E}_t(r_{i,t+1})$)：Var($z_{i,t}\beta^k$)

NN: Dropout+L2 NN →近似贝叶斯解释(附录证明)

L2：提供**高斯先验**对应的收缩项 ✚ **Dropout**: 前向传播时，某神经元以概率p停止工作→**随机正则化**→参数后验分布的近似抽样

Step 1：多次dropout，得到D个不同预测，计算预测方差 → **Step 2：**考虑载荷，从Λ的近似后验分布抽样计算方差

模拟检验

Design–Measure ex-ante forecast confidence

Test whether Lasso (NN) intervals provide proper uncertainty coverage under a linear (nonlinear) DGP:

Panel A: Probability that $CI_{\mu_{i,t}}$ contains the true simulated risk premiums $g(z_{i,t})$

Confidence Level	Lasso $CI_{\mu_{i,t}}$ Coverage	NN-3 $CI_{\mu_{i,t}}$ Coverage
10%	10.64%	9.61%
20%	21.17%	19.12%
30%	31.73%	28.73%
40%	42.14%	38.34%
50%	52.39%	48.03%
60%	62.60%	57.83%
70%	72.55%	67.59%
80%	82.20%	77.51%
90%	91.45%	87.54%
95%	95.74%	92.52%

Design–confidence H-L

1. The orthogonalized forecast error can be decomposed into 3 parts:

$$\mathbb{E}\left[\left(r_{i,t+1}-\Lambda_{i,t}f_{t+1}-\mathcal{F}(z^*_{i,t};\hat{\beta})\right)^2\right]=\underbrace{\left(\mathbb{E}(r_{i,t+1}-\Lambda_{i,t}f_{t+1})-\mathbb{E}(\mathcal{F}(z^*_{i,t};\hat{\beta}))\right)^2}_{\text{Bias}^2}$$
$$+\underbrace{\mathbb{E}\left[\left(\mathcal{F}(z^*_{i,t};\hat{\beta})-\mathbb{E}(\mathcal{F}(z^*_{i,t};\hat{\beta}))\right)^2\right]}_{\text{Variance}}+\underbrace{\text{Var}(\epsilon_{i,t+1})}_{\text{不可约误差}}$$

2. Economic intuition
   - **Bias**^{2}: model misspecification, whether model captures the true risk premium
   - **Variance**: parameter uncertainty, whether the forecast itself is stable
   - **Irreducible Error**: unpredictable shocks

3. ex ante forecast variance↑, ex post forecast errors↑, confi H-L portfolios rt ↑.

4. Under misspecification, Bias^2 dominates, reducing gains from confi H-L portfolios.

Design–Confidence H-L

Strategy	**Sorting Rule**	**Long / Short**
EW(VW) HL	Sort on next-month return forecasts	Long H, short L
EW(VW) Confident-HL	First sort on return forecasts, then on forecast confidence (EC)	Long $H10$, short $L10$
EW(VW) Low-Confident-HL	First sort on return forecasts, then on forecast confidence (EC)	Long $H1$, short $L1$
EW(VW) Low-Ivol	First sort on return forecasts, then on EC^{IVOL}	High-confidence, low-volatility subgroup
EW(VW) 1%-HL	Sort on return forecasts and select the most extreme 1%	Long top 1%, short bottom 1%

Design–Data and sample

- **Sample:** U.S. common stocks from CRSP, March 1957 to December 2020.
- **Coverage:** NYSE, AMEX, and NASDAQ stocks.
- **Predictors:** 176 predictors, including stock characteristics, industry dummies, and macro variables.
- **Split:** Training sample 1957–1974, validation sample 1975–1986, and out-of-sample test period 1987–2020.

Result–Main portfolio performance

		A: Equal-weighted strategies				*B: Value-weighted strategies*			
				FF-5+Mom				FF-5+Mom	
Model	Strategy	avg ret	*Sh*	α	*IR*	avg ret	*Sh*	α	*IR*
Lewellen	Confident-HL	2.52%	1.474	2.49%	1.844	1.85%	0.968	1.63%	0.993
	HL	1.24%	0.939	1.14%	0.989	0.58%	0.425	0.65%	0.564
	Low-Confident-HL	0.23%	0.128	0.27%	0.156	0.45%	0.249	0.54%	0.336
	1%-HL	1.31%	0.764	1.34%	0.856	0.66%	0.345	0.83%	0.471
	Low-Ivol-HL	1.21%	0.902	1.21%	1.049	0.98%	0.673	0.98%	0.735
Lasso	Confident-HL	3.48%	1.719	3.41%	1.703	2.51%	1.150	2.27%	1.057
	HL	1.92%	1.465	1.71%	1.444	0.89%	0.636	0.55%	0.607
	Low-Confident-HL	0.42%	0.207	0.17%	0.091	0.42%	0.223	0.12%	0.071
	1%-HL	2.28%	1.124	1.85%	1.038	1.34%	0.614	0.91%	0.510
	Low-Ivol-HL	1.32%	0.805	0.85%	0.614	0.99%	0.585	0.44%	0.338
NN-3	Confident-HL	4.56%	1.968	4.70%	2.112	3.21%	1.514	3.11%	1.569
	HL	2.76%	1.483	2.77%	1.558	1.84%	1.022	1.70%	1.055
	Low-Confident-HL	2.66%	1.036	2.76%	1.124	1.33%	0.446	1.70%	0.594
	1%-HL	3.81%	1.645	4.02%	1.787	2.15%	1.016	2.21%	1.100
	Low-Ivol-HL	2.89%	1.558	2.95%	1.653	2.40%	1.151	2.26%	1.189

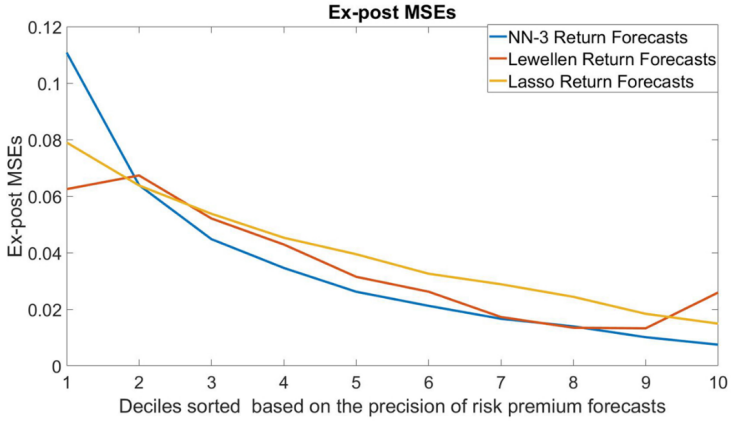
- The **Confident-HL** strategies consistently outperform the standard HL strategies

Result–Main portfolio performance

- The **Confident-HL** strategies outperform on complex models.

		Raw Returns		FF-5+Mom	
	Strategy	avg ret	Sh^2	α	IR^2
A: Lewellen model					
Confident-HL –	HL	**1.28%***	**1.29***	**1.34%***	**2.42***
		(0.000)	(0.000)	(0.001)	(0.000)
	Low-Confident-HL	**2.29%***	**2.16***	**2.22%***	**3.38***
		(0.000)	(0.000)	(0.000)	(0.000)
	1%-HL	**1.21%***	**1.59***	**1.15%***	**2.67***
		(0.000)	(0.000)	(0.001)	(0.000)
	Low-Ivol-HL	**1.31%***	**1.36***	**1.28%***	**2.30***
		(0.000)	(0.000)	(0.000)	(0.000)
B: Lasso model					
Confident-HL –	HL	**1.57%***	**0.81***	**1.70%***	**0.81***
		(0.000)	(0.000)	(0.000)	(0.000)
	Low-Confident-HL	**3.06%***	**2.91***	**3.23%***	**2.89***
		(0.000)	(0.000)	(0.001)	(0.000)
	1%-HL	**1.20%**	**1.69***	**1.55%***	**1.82***
		(0.022)	(0.000)	(0.000)	(0.000)
	Low-Ivol-HL	**2.16%***	**2.31***	**2.55%***	**2.52***
		(0.000)	(0.000)	(0.000)	(0.000)
C: NN-3 model					
Confident-HL –	HL	**1.79%***	**1.67***	**1.93%***	**2.03***
		(0.000)	(0.000)	(0.000)	(0.000)
	Low-Confident-HL	**1.90%***	**2.80***	**1.94%***	**3.20***

Result–Why does it work?



- lower ex-ante confidence → larger out-of-sample mean squared forecast errors.

复现-主要表格 1-模拟数据有效性检验

原文:		
Confidence Level	**Lasso Coverage**	**NN-3 Coverage**
10%	10.64%	9.61%
20%	21.17%	19.12%
30%	31.73%	28.73%
40%	42.14%	38.34%
50%	52.39%	48.03%
60%	62.60%	57.83%
70%	72.55%	67.59%
80%	82.20%	77.51%
90%	91.45%	87.54%
95%	95.74%	92.52%

复现:		
Confidence Level	**Lasso Coverage**	**NN-3 Coverage**
10%	10.83%	12.63%
20%	21.66%	24.97%
30%	32.44%	37.13%
40%	43.07%	48.93%
50%	53.51%	60.19%
60%	63.78%	70.78%
70%	73.63%	80.27%
80%	83.08%	88.60%
90%	92.07%	95.29%
95%	96.07%	97.86%

复现-主要表格 2-美国市场投资绩效

- 数据：1957-2016，94 特征
- 原文数据：1957-2020，94 特征 + 宏观 + 行业 =176

	原文	EW		VW		VW		VW	
Model	**Strategy**	**ret**	*Sh*	**ret**	*Sh*	**ret**	*Sh*	**ret**	*Sh*
Lewellen	Confit-HL	2.29%	1.10	1.41%	0.66	2.43%	0.89	1.27%	0.47
	HL	1.30%	0.87	0.40%	0.31	1.73%	0.92	0.00%	-0.03
	Low-Confi-HL	0.34%	0.17	0.32%	0.15	1.24%	0.61	-0.15%	-0.08
Lasso	Confi-HL	4.46%	1.80	1.63%	0.60	3.41%	2.05	2.22%	1.23
	HL	2.03%	1.45	0.65%	0.45	2.97%	2.34	0.59%	0.44
	Low-Confit-HL	0.77%	0.45	0.64%	0.36	2.17%	1.52	0.11%	0.07
NN-3	Confi-HL	4.82%	1.94	2.88%	1.12	1.43%	0.65	1.61%	0.58
	HL	2.79%	1.70	0.93%	0.57	1.43%	0.81	2.48%	1.33
	Low-Confit-HL	1.63%	0.69	0.95%	0.40	2.04%	1.00	3.51%	1.36

复现-主要表格 2-中国市场投资绩效

- 数据：1997-2024，94 特征
- 原文数据：1957-2020，94 特征 + 宏观 + 行业 =176

	原文	EW		VW		VW		VW	
Model	**Strategy**	**ret**	*Sh*	**ret**	*Sh*	**ret**	*Sh*	**ret**	*Sh*
Lewellen	Confit-HL	2.29%	1.10	1.41%	0.66	0.56%	0.35	0.50%	0.27
	HL	1.30%	0.87	0.40%	0.31	0.51%	0.37	0.45%	0.23
	Low-Confi-HL	0.34%	0.17	0.32%	0.15	0.47%	0.27	0.26%	0.14
Lasso	Confi-HL	4.46%	1.80	1.63%	0.60	1.64%	0.89	1.42%	0.70
	HL	2.03%	1.45	0.65%	0.45	0.77%	0.60	0.03%	0.03
	Low-Confit-HL	0.77%	0.45	0.64%	0.36	0.04%	0.03	-0.22%	-0.14
NN-3	Confi-HL	4.82%	1.94	2.88%	1.12	5.70%	0.49	2.10%	0.16
	HL	2.79%	1.70	0.93%	0.57	5.84%	0.45	2.26%	0.15
	Low-Confit-HL	1.63%	0.69	0.95%	0.40	6.00%	0.42	1.26%	0.09

Thanks!

Question & idea?

Idea

- 参数不确定性 v.s. 模型不确定性 v.s. 信息不确定性（分歧）
- 不确定性来自横截面 or 时序?
- 考虑不确定性的投资组合优化
- 预测时考虑其他数据结构，如预测分布，如在债券市场考虑尾部风险：predicted VaR、ES
- 从“个股”扩展到“组合 / 因子 / 行业 / 市场层面”

Appendix 1：线性模型收益预测与残差结构 I

- **收益预测方程:**

$$r_{i,t+1} = z_{i,t}^{\top}\beta + \eta_{i,t+1}$$

  - $r_{i,t+1}$ 表示股票 i 在 $t+1$ 的超额收益；$z_{i,t}$ 表示时点 t 的股票特征变量；β 为待估计参数，$\eta_{i,t+1}$ 为模型没有解释掉的残差

- **然而，需要重视：残差不是彼此独立的，存在截面相关性**

$$\eta_{i,t+1} = \Lambda_{i,t} f_{t+1} + \varepsilon_{i,t+1}$$

现实里很多股票会一起受到市场影响，比如宏观数据公布等，所以某只股票预测错了，往往不是它单独错，很多股票可能一起错。因此残差 = 共同市场冲击 + 个股自己的噪声。其中，共同因子 f_{t+1} 用于刻画市场层面的共同冲击 (以市场超额收益的意外变化刻画 $Mkt_{t+1} - E_t(Mkt_{t+1})$)，$\varepsilon_{i,t+1}$ 为个体特有噪声。

Appendix 1：线性模型收益预测与残差结构 II

$\Lambda_{i,t}$ 为股票 i 对这个共同冲击的敏感度，不是固定不变，而是由股票特征决定：

$$\Lambda_{i,t} = a_0 + a_1 Size_{i,t} + a_2 BM_{i,t} + a_3 Mom_{i,t}$$

- $Mom_{i,t}$ 是企业 i 在时间 t 的 1 年动量回报；
- a0,a1,a2,a3 是未知的载荷系数，先验值将在下一小节中指定
- 优点：既允许载荷随特征变化，又能处理短历史样本股票的估计不稳问题

Appendix 2：线性模型收益预测的不确定性 (1) I

❶ 估计参数 β，但不是普通的 penalized MSE

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{N^{Tr}N^{S}} \sum_{t\in Tr}\sum_{i\in S} \left(r_{i,t+1} - z_{i,t}^{\top}\beta - \Lambda_{i,t}f_{t+1}\right)^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2$$

$$\hat{\beta} = \arg\min_{\beta} \frac{1}{N^{Tr}N^{S}} \sum_{t\in Tr}\sum_{i\in S} \left(\bar{r}_{i,t+1} - z_{i,t}^{\top}\beta\right)^2 + \lambda_1\|\beta\|_1 + \lambda_2\|\beta\|_2$$

- 其中，$\bar{r}_{i,t+1} = r_{i,t+1} - \Lambda_{i,t}f_{t+1}$ 为**正交化收益**。
- 线性回归：$\lambda_1 = 0,\ \lambda_2 = 0$; Lasso：$\lambda_1 \neq 0,\ \lambda_2 = 0$; Ridge：$\lambda_1 = 0,\ \lambda_2 \neq 0$; Elastic Net：$\lambda_1 \neq 0,\ \lambda_2 \neq 0$
- 为后续贝叶斯解释提供基础; 进一步用于计算预测方差与置信区间

Appendix 2：线性模型收益预测的不确定性 (1) II

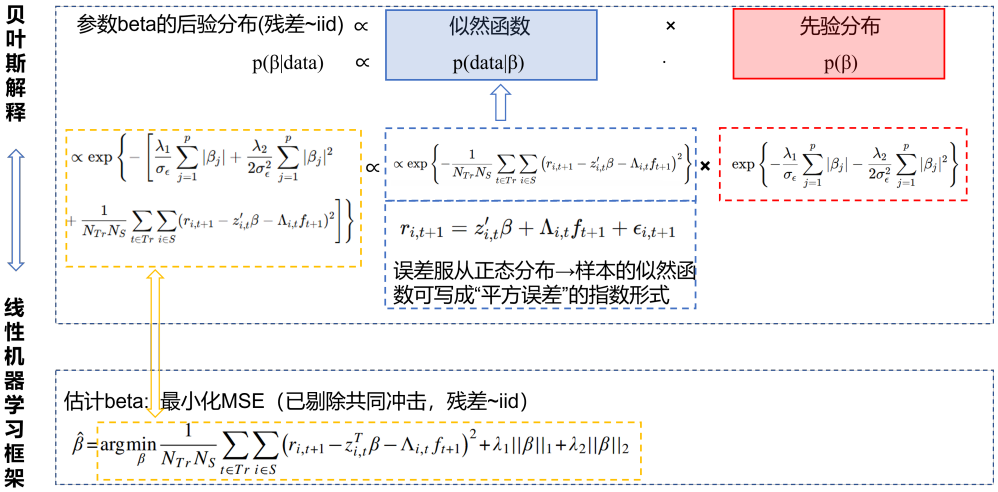
2. **β 感兴趣参数的贝叶斯解释**

定理 1：当设置先验：double exponential prior on β（红色框），模型的对数似然估计等价于估计参数的后验分布

$$\pi(\beta \mid \lambda_1, \lambda_2, \sigma_\epsilon^2) \propto \exp\left\{ -\frac{\lambda_1}{\sigma_\epsilon} \sum_{j=1}^{P} |\beta_j| - \frac{\lambda_2}{2\sigma_\epsilon^2} \sum_{j=1}^{P} |\beta_j|^2 \right\}$$

此过程需要同时估计其他未知参数的后验，例如正则化超参数（λ_1 和 λ_2 ）和因子载荷（α_0 、α_1 、α_2 和 α_3 ）。下面概述了联合估计所有这些参数的贝叶斯框架。

Appendix 2：线性模型收益预测的不确定性 (1) III



Appendix 2：线性模型收益预测的不确定性 (2)

❸ 设置先验

参数	处理方式	先验公式	直觉
β	双指数先验	$\pi(\beta) \propto \exp\Big(-\frac{\lambda_1}{\sigma}\sum\lvert\beta_j\rvert - \frac{\lambda_2}{2\sigma^2}\sum\beta_j^2\Big)$ $\pi(\beta \mid \lambda_1, \lambda_2, \sigma_\epsilon^2, \tau_1^2, \ldots, \tau_P^2) \sim N_P\big(0_P, \sigma_\epsilon^2 D_\tau\big)$,	Lasso/Ridge
β	引入辅助变量 τ_j^2，转为正态分布	$D_\tau = \mathrm{diag}\Big((\tau_1^{-2}+\lambda_2)^{-1}, \ldots, (\tau_P^{-2}+\lambda_2)^{-1}\Big)$, $\tau_1^2, \ldots, \tau_P^2 \sim \prod_{j=1}^{P} \frac{\lambda_1^2}{2}\exp\big(-\lambda_1^2\tau_j^2/2\big)\, d\tau_j^2$,	正态，可用 Gibbs Sampling
σ^2	Jeffreys 先验	$\pi(\sigma^2) = 1/\sigma^2$	弱信息
λ_1	Gamma	$\lambda_1^2 \sim \mathrm{Gamma}(r_1, \delta_1)$	不再 ML 调参
λ_2	Gamma	$\lambda_2 \sim \mathrm{Gamma}(r_2, \delta_2)$	λ>0,Kyung(2010)
$a_0 \sim a_3$	正态	$(a_0, \ldots, a_3)^\top \sim N\Big(\nu_1(X_e^\top X_e)^{-1}X_e^\top R_e,$ $\nu_2(X_e^\top X_e)^{-1}\Big)$	共轭正态分布

Appendix 2：线性模型收益预测的不确定性 (3) I

4. **估计所有参数的联合后验分布**
   - 在给定式 (8)–(14) 中的先验分布后，模型参数的联合后验分布通常无法直接写成闭式解，因此本文采用 **Gibbs 抽样**进行估计。
   - **基本思想：逐个抽取条件后验分布**
     - $\pi(\theta \mid \Theta_{\sim\theta}, Z, R)$：参数 θ 的条件后验分布
     - 每一步固定其他参数，只对某一个参数进行抽样
     - 条件信息包括：其他参数的最新取值、训练样本数据 (训练样本中堆叠后的特征矩阵 Z、超额收益向量 R)
     - 不断重复后，抽样序列收敛于参数的真实联合后验分布

Appendix 2：线性模型收益预测的不确定性 (3) II

5 利用后验分布估计预测方差，从而计算置信水平

- 利用每一次 Gibbs 抽样得到的参数 β^k，可形成一组预测值 $\{z_{i,t}^{*\top}\beta^k\}$，其离散程度定义为风险溢价预测方差：

$$\widehat{\mathrm{Var}}\left(\hat{E}_t(r_{i,t+1}^*)\right) = \frac{1}{N^{\mathrm{Gibbs}}}\sum_{k=1}^{N^{\mathrm{Gibbs}}}\left(z_{i,t}^{*\top}\beta^k - z_{i,t}^{*\top}\bar{\beta}\right)^2, \bar{\beta} = \frac{1}{N^{\mathrm{Gibbs}}}\sum_{k=1}^{N^{\mathrm{Gibbs}}}\beta^k$$

- **原始收益预测方差**：原始超额收益的预测方差进一步分解为：

$$\widehat{\mathrm{Var}}(r_{i,t+1}^*) = \widehat{\mathrm{Var}}\left(\hat{E}_t(r_{i,t+1}^*)\right) + \hat{\Lambda}_{i,t}^2\hat{\sigma}_f^2 + \hat{\sigma}_\epsilon^2$$

即同时考虑：

- 参数后验不确定性; 因子波动带来的系统性风险; 特质噪声带来的剩余不确定性

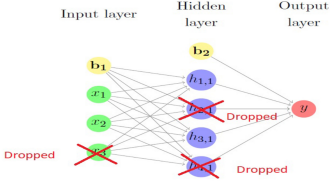
Appendix 2：线性模型收益预测的不确定性 (3) III

6 模拟检验

- 结果表明，模型构造的风险溢价置信区间和原始收益预测带具有较好的覆盖率
- 例如，95% 区间大致能以 95% 的概率覆盖真实值。
- 这说明本文不仅能预测收益，还能较准确地衡量预测不确定性。

Appendix 3：神经网络收益预测的不确定性 (1)

Dropout：在前向传播的时候，让某个神经元的激活值以一定的概率(1− p)停止工作



dropout mask →NN输出也随机→**随机正则化**

→可以解释成"来自参数后验分布的近似抽样

L2：提供**高斯先验**对应的收缩项

高斯先验 ⟺ L2 惩罚

Dropout+L2 NN →近似贝叶斯解释

$$\hat{\beta}_{\Lambda_{i,t},\lambda_2,p} = \arg\min_{\beta} \frac{\sum_{t\in Tr}\sum_{i\in S}\left(r_{i,t+1}-\Lambda_{i,t}f_{t+1}-\left(b_2+\phi\left(b_1+z_{i,t}^{T}(p_{1it}W_1)\right)(p_{2it}W_2)\right)\right)^2}{N_{Tr}N_S} + \lambda_2\left(\|W_1\|^2+|0W_2\|^2+\|b_1\|^2+\|b_2\|^2\right), \quad (28)$$

β：W1和W2是连接输入层到隐藏层、隐藏层到输出层的权重矩阵，截距 b1 和 b2 为对应层截距

点预测+置信度预测

Appendix 3：神经网络收益预测的不确定性 (2)

Dropout+L2 NN →近似贝叶斯解释

$$\hat{\beta}_{\Lambda_{i,t},\lambda_2,p} = \arg\min_{\beta} \frac{\sum_{t\in Tr}\sum_{i\in S}\left(r_{i,t+1}-\Lambda_{i,t}f_{t+1}-\left(b_2+\phi\left(b_1+z_{i,t}^{T}(p_{1it}W_1)\right)(p_{2it}W_2)\right)\right)^2}{N_{Tr}N_S} + \lambda_2\left(\|W_1\|^2+|0W_2\|^2+\|b_1\|^2+\|b_2\|^2\right), \quad (28)$$

β：W1和W2是连接输入层到隐藏层、隐藏层到输出层的权重矩阵，
截距 b1 和 b2 为对应层截距

点预测：给定 Λ，神经网络对股票 i下一期风险溢价的预测--不 dropout

置信度预测：

Step 1：多次dropout，得到 D个不同预测，计算出预测方差：

$$\widehat{\mathrm{Var}}(E^{*}_{\Lambda_{i,t},\lambda_2,p}) = \frac{1}{D}\sum_{d=1}^{D}\left(E^{*}_{\Lambda_{i,t},\lambda_2,p,d} - \overline{E^{*}}\right)^2$$

Step 2：但现实里Λ未知，需要考虑其后验不确定性，从Λ的近似后验分布里抽取 S_{NN} 个样本。

$$\widehat{\mathrm{Var}}(\hat{E}_t(r_{i,t+1})) = \frac{1}{S_{NN}}\sum_{s=1}^{S_{NN}}\widehat{\mathrm{Var}}(E^{*}_{\Lambda^{s}_{i,t},\lambda_2,p}) + \frac{1}{S_{NN}}\sum_{s=1}^{S_{NN}}\left(E^{*}_{\Lambda^{s}_{i,t},\lambda_2,p} - E^{*,Dropout}_{it,\lambda_2,p}\right)^2$$

神经网络自身预测的不确定性
——来自测试时重复 dropout

不同 Λ 取值带来的额外不确定性
——来自因子载荷估计误差